

Invention Disclosure IDF-205682

Submitted by Wenfeng Wang on Apr 13 2026

Title:	RRM-Driven, Radio-Aware Automated Frequency Coordination (AFC) Request Management and Enforcement for Multi-Radio Access Points
Problem:	Existing AFC implementations focus primarily on geolocation, regulatory compliance, and channel/power authorization at an access-point or band level. In modern deployments, APs often contain heterogeneous combinations of dedicated and shared radios capable of operating on overlapping bands. Additionally, AFC integration requires robust handling of provider selection, retries, expiry, location changes, and regulatory edge cases. Without RRM-driven coordination, AFC workflows suffer from inefficiencies, delayed recovery, improper power selection, incorrect band targeting, and inability to adapt to radio topology, AP model, or operational state changes. Current solutions do not sufficiently leverage RRM context to determine when, how, and from which radio AFC requests should be issued, validated, retried, or re-applied.
Impact:	
Technical Context:	The invention relates to Automated Frequency Coordination (AFC) in wireless local area networks, and more specifically to radio-aware lifecycle management of AFC requests and responses driven by radio resource management (RRM) in multi-radio access points (APs) operating across 2.4GHz, 5GHz, and 6GHz bands.
Solution Description:	This invention introduces a comprehensive RRM-driven AFC control framework that governs the full AFC lifecycle—including request generation, provider selection, response validation, retry, expiry handling, and radio-specific enforcement—on a per-AP, per-band, and per-radio basis. RRM logic determines AFC request parameters, issues AFC POST/GET requests, interprets AFC responses, and applies channel, EIRP, and PSD constraints to individual radios. The framework ensures resilient AFC behavior across AP models, bands (including non-UNII 5/7 channels), and dynamic operational conditions such as AP movement, radio failures, and multi-provider deployments.
Previous Solution Comparison:	
Uniqueness:	Key Innovation Elements • RRM-Driven AFC Request Construction RRM determines parameters required for AFC requests, including ruleset ID, certificate ID (based on AP model and country), representative channel mappings (including 320-1 / 320-2), and radio eligibility. • Radio-Aware AFC Request Origination For dual-band and tri-band APs with 6GHz capability, RRM selectively sends AFC requests from the appropriate band (including initiating requests from 2.4GHz radios when required), even when a dedicated 6GHz radio is present. • Dynamic Provider Selection and Redirection RRM logic supports redirecting AFC requests to different AFC providers based on policy, availability, or region. • Robust Retry and Validation Mechanisms RRM retries AFC POST operations when NULL channel information is returned and validates cached AFC responses using GET requests, ensuring consistency and correctness. • Radio-Specific Response Interpretation RRM selects the best interference-free channel and corresponding EIRP/PSD values from AFC responses and applies tx-power constraints by mapping transmit power below channel EIRP limits. • Lifecycle State Management AP claim and unclaim workflows are managed by RRM,

including clearing GPS data from cache, waiting for initial GPS stabilization (e.g., first five minutes), issuing AFC POST requests, and handling location transitions. • Expiry and Location Change Handling RRM enforces a defined AFC expiry interval (e.g., 16hours), validates cached responses, and re-initiates AFC workflows upon detected AP location changes. • Regulatory Edge-Case Handling RRM logic handles non-UNII 5/7 channels and ensures compliant operation across regulatory domains.

Usage Details: Radio Resources Management
Projects or Products: Radio Resources Management
Solution Detectability:
Third Party Involvement:
Third Party Involvement
Details:
External Disclosure Date:
Technology: a065Y00001vvjg9QAA
Secondary Technologies:

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Attachments